Generalized Pandrosion Residuals

Residual profile of an iterative geometric construction for p-th roots of any positive real number

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Abstract

We revisit Pandrosion's classical geometric construction, reported by Pappus, for approximating the cube root of two and generalize it to the computation of p^{th} roots of any positive real x by inserting proportional means using Thales' theorem. In our formulation the construction reduces to a single parameter $s_n \in (0,1)$ together with an output v_n converging to $x^{1/p}$. After a normalization the iteration takes a universal form involving the geometric sum

$$S_p(s) = 1 + s + \cdots + s^{p-1}.$$

We identify the unique fixed point $s^* = x^{-1/p}$, show that the convergence is *linear*, and derive an explicit closed-form expression for the associated contraction ratio

$$\lambda_{p,x} = \frac{(\alpha - 1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p}.$$

This ratio depends only on the target α; throughout the paper we refer to it as the *convergence signature* of Pandrosion's method, a memorable shorthand for what is formally the standard asymptotic error constant $h'(s^*)$ of a linearly convergent fixed-point iteration. Building on this formula we obtain non-asymptotic error bounds, prove that $\lambda_{p,x}$ is monotonic in both the order p and the argument x, extend the construction to the case $0 < x < 1$ (where convergence becomes oscillatory), and determine an optimal starting point that minimizes the initial error. We further show that Steffensen's acceleration applied to the Pandrosion iteration yields *quadratic* convergence, and we propose a simple scaling strategy that exploits the monotonicity of $\lambda_{p,x}$ to reduce the computational cost for large x. Taken together, these results situate an ancient geometric procedure within modern fixed-point theory and suggest derivative-free root-finding algorithms inspired by Pandrosion.

Contents

1 Introduction

The Greek problem of doubling the cube consists of constructing a segment of length $\sqrt[3]{2}$ from a unit segment. It is equivalent to finding two proportional means m_1, m_2 between 1 and 2, that is,

$$\frac{1}{m_1} = \frac{m_1}{m_2} = \frac{m_2}{2},$$

which yields $m_1 = \sqrt[3]{2}$ and $m_2 = \sqrt[3]{4}$. Pandrosion of Alexandria proposed an *approximate* geometric method, reported by Pappus in the *Collectio*, Book III [3]: by successive parallels (Thales' theorem), she constructs proportional ratios that approach these means without ever reaching them exactly in a finite number of steps.

This work starts from a fundamental observation:

> *Mathematically the target number is exact, but any approximation procedure yields a measurable residual profile — a footprint intrinsic to the chosen method.*

At each finite step of Pandrosion's construction, there remains a gap $\varepsilon_n = |v_n - \alpha|$ between the geometric approximation and the ideal value. The sequence (ε_n) is the **residual profile** of the method. Our goal is to compute this profile exactly for any p-th root $x^{1/p}$, and to identify the contraction ratio that characterizes Pandrosion's method.

Positioning and related work. Rational iterations for p-th roots are a classical topic with a large literature: Newton's method applied to $f(u) = u^p - x$, Halley's method, and more generally Schröder's (1870) and König's (1884) families produce iterations of arbitrary order with explicit asymptotic constants. The systematic analysis of fixed-point iterations and their acceleration goes back to Ostrowski [9], Traub [11], and Householder [10]; the derivative-free counterpart via Aitken–Steffensen (1926/1933) is standard and appears in every modern numerical-analysis textbook [5, 6]. What is specific to the iteration (3) below is that it arises directly from Pandrosion's geometric construction, *without* being derived as Newton or Schröder applied to a particular polynomial: it is a rational iteration with linear (not quadratic) order, whose contraction ratio admits the closed form (5) as a rational function of $\alpha = x^{1/p}$. We have not located this closed form, together with its monotonicity in (p, x) and its use to drive a scaling preconditioning, in the textbooks [5, 6] or the historical literature on cube duplication, but we make no claim that the iteration itself is genuinely new: the contribution is to record the profile of a specific historical iteration and to show how standard acceleration devices (Aitken Δ^2, scaling) interact with it.

Outline. Section 2 recalls Pandrosion's construction for $\sqrt[3]{2}$ and analyzes its residual profile. Section 4 states the generalization to any p, x. Section 5 computes the contraction ratio $\lambda_{p,x}$ and provides closed-form expressions. Section 7 presents numerical tables. Section 8 compares with bisection, the secant method, and Newton's method. Section 9 establishes functional properties of $\lambda_{p,x}$. Section 10 provides non-asymptotic error bounds. Section 11 extends the theory to $x < 1$. Section 12 determines the optimal starting point. Section 13 shows that Steffensen's acceleration of Pandrosion achieves quadratic convergence. Section 14 exploits the monotonicity of $\lambda_{p,x}$ to optimize the method for large x via scaling. Section 15 concludes.

2 Pandrosion's construction for $\sqrt[3]{2}$

2.1 The two sequences

The geometric construction produces two sequences as a function of an initial parameter u_0:

$$u_{n+1} = \frac{u_n}{2 - 2\left(\dfrac{4 - u_n}{4}\right)^3}, \qquad v_n = 2\left(\frac{4 - u_n}{4}\right)^2, \qquad u_0 = 0.5. \tag{1}$$

The sequence (u_n) is the *internal parameter* of the construction (a length in the Thales diagram). The sequence (v_n) is the *readout*: the approximation of $\sqrt[3]{2}$ produced at step n.

Proposition 2.1. $\lim_{n\to\infty} v_n = \sqrt[3]{2}$.

2.2 Normalized substitution

Setting $s_n = (4 - u_n)/4$, so that $s_n \in (0, 1)$, we obtain:

- $u_n = 4(1 - s_n)$,
- $v_n = 2s_n^2$,
- $u_{n+1} = \dfrac{4(1 - s_n)}{2(1 - s_n^3)} = \dfrac{2}{1 + s_n + s_n^2}$.

The iteration on s_n then becomes:

$$s_{n+1} = 1 - \frac{u_{n+1}}{4} = \frac{2s_n^2 + 2s_n + 1}{2s_n^2 + 2s_n + 2}. \tag{2}$$

2.3 Fixed point

The fixed point s^* satisfies $s^* = (2s^{*2}+2s^*+1)/(2s^{*2}+2s^*+2)$, which gives $2(1-s^*)(1+s^*+s^{*2}) = 1$, i.e. $2(1 - s^{*3}) = 1$, hence $s^{*3} = 1/2$, so:

$$s^* = 2^{-1/3} = \frac{1}{\sqrt[3]{2}}.$$

The output at the fixed point is $v^* = 2s^{*2} = 2 \cdot 2^{-2/3} = 2^{1/3} = \sqrt[3]{2}$. ✓

2.4 Residual profile: linear convergence

Let $h(s) = (2s^2 + 2s + 1)/(2s^2 + 2s + 2)$. The derivative at s^* gives the convergence rate:

$$h'(s^*) = \frac{4s^* + 2}{(2s^{*2} + 2s^* + 2)^2} = \frac{2(2s^* + 1)}{4(s^{*2} + s^* + 1)^2} = \frac{2s^* + 1}{2(s^{*2} + s^* + 1)^2}.$$

Substituting $s^* = 2^{-1/3}$ and using the closed form (7):

$$\lambda_{3,2} = \frac{(\sqrt[3]{2} - 1)(\sqrt[3]{2} + 2)}{\sqrt[3]{4} + \sqrt[3]{2} + 1} \approx 0.2202.$$

Theorem 2.1 (Linear residual law for $p = 3$, $x = 2$)**.** Under the above hypotheses, for any s_0 sufficiently close to s^*:

$$\varepsilon_{n+1}^{(s)} \approx \lambda_{3,2}\, \varepsilon_n^{(s)}, \qquad \lambda_{3,2} \approx 0.2202.$$

For the output, a Taylor expansion of $v(s) = xs^{p-1}$ at $s^* = \alpha^{-1}$ gives $v_n - \alpha \approx x(p - 1)(s^*)^{p-2}(s_n - s^*)$. Using $x(s^*)^{p-2} = \alpha^p \cdot \alpha^{-(p-2)} = \alpha^2$, this rewrites as $v_n - \alpha \approx (p-1)\alpha^2(s_n - s^*)$; the two forms are equal. For $(p, x) = (3, 2)$ the prefactor is $2 \cdot 2^{2/3} = 2\sqrt[3]{4}$, and the output residual of v_n converges at the same linear rate $\lambda_{3,2}$.

3 Geometric origin: why S_p arises naturally

3.1 The construction by successive parallels

Pandrosion's construction relies on Thales' theorem applied to a rectangle. Consider a rectangle with sides H and W equipped with two diagonals (Figure 1). A horizontal line at height u_n from the bottom intersects the two diagonals and defines:

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt[3]{2}\rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.8750	1.5312	2.713×10^{-1}	—
1	0.8107	1.3143	5.44×10^{-2}	0.200
2	0.7974	1.2717	1.17×10^{-2}	0.216
3	0.7945	1.2625	2.58×10^{-3}	0.219
4	0.7939	1.2605	5.67×10^{-4}	0.220
∞	$2^{-1/3}$	$\sqrt[3]{2}$	0	$\to \lambda_{3,2}$

Table 1: Numerical residual profile for $p = 3$, $x = 2$ ($u_0 = 0.5$, i.e. $s_0 = 0.875$). The ratio converges to $\lambda_{3,2} \approx 0.220$.

- the parameter $s_n = (H - u_n)/H$ (normalized position from the top),
- the readout length $v_n = x \cdot s_n^{p-1}$ (the output of the construction).

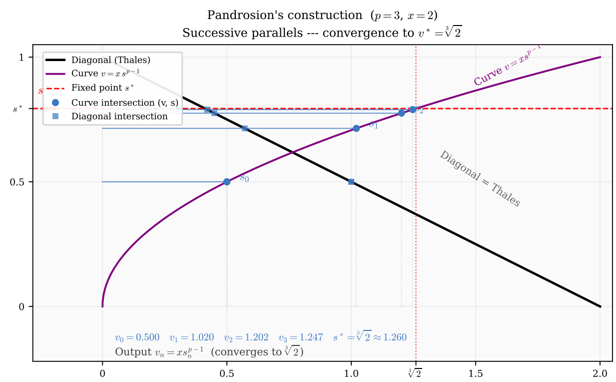

Figure 1: Pandrosion's geometric construction for $p = 3$, $x = 2$. The blue horizontal lines (steps $n = 0, 1, 2, 3$) converge toward the dashed red line, which corresponds to $v^* = \sqrt[3]{2}$.

3.2 Where does the polynomial S_p come from?

The fundamental idea is as follows. We seek to insert $p - 1$ proportional means between 1 and x, that is, to construct the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$ where $\alpha = x^{1/p}$. If $s_n \approx \alpha^{-1}$ is the current approximation of the inverse ratio, the sum of the approximate geometric sequence is:

$$S_p(s_n) = 1 + s_n + s_n^2 + \cdots + s_n^{p-1}.$$

The ideal sequence satisfies $x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1$, which rewrites as $x(1 - s^{*p}) = x - 1$. Pandrosion's correction adjusts s_n so as to reduce the gap between $S_p(s_n)$ and its ideal value via the formula:

$$s_{n+1} = 1 - \frac{x - 1}{x \cdot S_p(s_n)}.$$

Geometrically, the ratio $(x - 1)/(x \cdot S_p(s_n))$ measures the *deviation* of the current proportional chain from the perfect geometric chain, and subtraction from 1 produces the corrected ratio.

The polynomial S_p is thus the analytical transcription of the sum of the p segments constructed via Thales' theorem.

4 Generalization: Pandrosion's construction for $x^{1/p}$

4.1 The universal iteration

Let $x > 0$ and $p \geq 2$ be an integer. Set $\alpha = x^{1/p}$. The generalized Pandrosion geometric construction is defined as follows.

Definition 4.1 (Generalized Pandrosion iteration)**.** We define the sequence $(s_n)_{n\geq 0}$ in $(0,1)$ by

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k = \frac{1-s^p}{1-s}, \tag{3}$$

and the output sequence by

$$v_n = x \cdot s_n^{p-1}. \tag{4}$$

We call $\varepsilon_n = |v_n - \alpha|$ the **Pandrosion residual** at step n.

Remark 4.1. For $p = 3$, $x = 2$, we recover exactly the iteration (2). Indeed:

$$1 - \frac{1}{2(1+s+s^2)} = \frac{2s^2+2s+1}{2s^2+2s+2}.\checkmark$$

4.2 Geometric interpretation

The iteration (3) analytically translates the Thales operation: at each step, $S_p(s_n)$ measures the "geometric sum" of the constructed segments, and the correction $(x-1)/(x \cdot S_p(s_n))$ adjusts the position proportionally to the gap with the ideal geometric sequence $1, \alpha, \alpha^2, \ldots, \alpha^{p-1}, x$.

4.3 Fixed point

Theorem 4.1 (Fixed point)**.** The unique fixed point of the iteration (3) in $(0,1)$ is $s^* = x^{-1/p} = \alpha^{-1}$. The corresponding output is $v^* = x \cdot (x^{-1/p})^{p-1} = x^{1/p} = \alpha$.

Proof. At a fixed point s^*, we have $1 - (x-1)/(x \cdot S_p(s^*)) = s^*$, i.e.

$$x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1.$$

Since $S_p(s)(1-s) = 1 - s^p$, this gives $x(1 - s^{*p}) = x - 1$, hence $s^{*p} = 1/x$, i.e. $s^* = x^{-1/p}$. The formula for v^* follows immediately. □

5 The contraction ratio $\lambda_{p,x}$

5.1 General computation

Theorem 5.1 (Pandrosion contraction ratio)**.** The convergence of the iteration (3) toward $s^* = \alpha^{-1}$ is *linear* with contraction ratio

$$\boxed{\lambda_{p,x} = \frac{(\alpha-1)\displaystyle\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\displaystyle\sum_{k=0}^{p-1} \alpha^k}}, \qquad \alpha = x^{1/p}. \tag{5}$$

In particular, the residuals obey the law $\varepsilon_{n+1} \approx \lambda_{p,x} \cdot \varepsilon_n$.

Proof. Let $h(s) = 1 - (x-1)/(x \cdot S_p(s))$. Then

$$h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2},$$

where $S_p'(s) = \sum_{k=1}^{p-1} k s^{k-1}$.

At the fixed point $s^* = \alpha^{-1}$:

$$S_p(s^*) = \frac{1-\alpha^{-p}}{1-\alpha^{-1}} = \frac{\alpha(x-1)}{x(\alpha-1)},$$

$$S_p'(s^*) = \sum_{k=1}^{p-1} k\,\alpha^{-(k-1)} = \alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Substituting:

$$h'(s^*) = \frac{x-1}{x} \cdot \frac{\alpha \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\left[\dfrac{\alpha(x-1)}{x(\alpha-1)}\right]^2} = \frac{x(\alpha-1)^2}{\alpha(x-1)} \sum_{k=1}^{p-1} k\,\alpha^{-k}.$$

Multiplying numerator and denominator by α^{p-1}, and using $x = \alpha^p$:

$$h'(s^*) = \frac{(\alpha-1)^2\,\alpha^{p-1} \sum_{k=1}^{p-1} k\,\alpha^{-k}}{\alpha^p - 1} = \frac{(\alpha-1) \sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k},$$

where we used $\alpha^p - 1 = (\alpha-1)\sum_{k=0}^{p-1} \alpha^k$. This establishes (5). □

Remark 5.1. The contraction ratio $\lambda_{p,x}$ is entirely determined by $\alpha = x^{1/p}$: the target itself dictates the rate at which Pandrosion's method approaches it. We will occasionally refer to $\lambda_{p,x}$, viewed as a rational function of α, as the *convergence signature* of the method. This is purely terminological — $\lambda_{p,x}$ is of course nothing more than $h'(s^*)$, the standard asymptotic error constant of a linearly convergent fixed-point iteration — and we introduce the name only to make the dependence on the target (rather than on the starting point) verbally prominent.

5.2 Closed forms for $p = 2$ and $p = 3$

For $p = 2$: $\sum_{k=1}^{1} k\,\alpha^0 = 1$, $\sum_{k=0}^{1} \alpha^k = 1 + \alpha$.

$$\lambda_{2,x} = \frac{\alpha-1}{\alpha+1} = \frac{\sqrt{x}-1}{\sqrt{x}+1}. \tag{6}$$

For $p = 3$: $\sum_{k=1}^{2} k\,\alpha^{2-k} = \alpha + 2$, $\sum_{k=0}^{2} \alpha^k = \alpha^2 + \alpha + 1$.

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2+\alpha+1} = \frac{(\sqrt[3]{x}-1)(\sqrt[3]{x}+2)}{\sqrt[3]{x^2}+\sqrt[3]{x}+1}. \tag{7}$$

For $p = 4$: $\sum_{k=1}^{3} k\,\alpha^{3-k} = \alpha^2 + 2\alpha + 3$, $\sum_{k=0}^{3} \alpha^k = \alpha^3 + \alpha^2 + \alpha + 1$.

$$\lambda_{4,x} = \frac{(\alpha-1)(\alpha^2+2\alpha+3)}{\alpha^3+\alpha^2+\alpha+1}. \tag{8}$$

Remark 5.2 (Asymptotic behavior in x)**.** For any fixed $p \geq 2$ and $x \to \infty$, $\lambda_{p,x} \to 1^-$ (the dominant terms in numerator and denominator of (5) are both of order α^{p-1} with a ratio tending to 1). The larger x, the slower the convergence. For $x \to 1^+$, $\lambda_{p,x} \to 0$ by Proposition 9.1(1): convergence is very fast near 1.

5.3 Basin of attraction

Theorem 5.2 (Global convergence)**.** For any $x > 1$ and any $p \geq 2$, the iteration (3) converges to s^* for all $s_0 \in (0,1)$. Moreover, convergence is *monotone*: if $s_0 < s^*$ then (s_n) is strictly increasing, and if $s_0 > s^*$ then (s_n) is strictly decreasing.

Proof. Set $h(s) = 1 - (x-1)/(x \cdot S_p(s))$.

Step 1: h **is strictly increasing.** We have $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$, where $S_p'(s) = \sum_{k=1}^{p-1} ks^{k-1} > 0$ for $s > 0$.

Step 2: h **maps** $(0,1)$ **into itself.** For $s \in (0,1)$, $S_p(s) > 1$ since the first term equals 1 and the remaining terms are strictly positive. Hence $(x-1)/(x\, S_p(s)) < (x-1)/x < 1$, which gives $h(s) > 0$. On the other hand $(x-1)/(x\, S_p(s)) > 0$ for $x > 1$, so $h(s) < 1$. Thus $h((0,1)) \subset (0,1)$. (At the boundary: $h(0^+) = 1 - (x-1)/x = 1/x \in (0,1)$ and $h(1^-) = 1 - (x-1)/(xp) \in (0,1)$.)

Step 3: uniqueness of the fixed point on $(0,1)$**.** The fixed-point equation $h(s) = s$ rewrites as $x\, S_p(s)(1-s) = x - 1$, i.e. $x(1 - s^p) = x - 1$, i.e. $s^p = 1/x$. On $(0,1)$ this has the unique solution $s^* = x^{-1/p}$. In particular, the sign of $h(s) - s$ is constant on each of $(0, s^*)$ and $(s^*, 1)$.

Step 4: the sign of $h(s) - s$**.** At $s = 1/x \in (0, s^*)$ (using $x > 1$ and $s^* = x^{-1/p} > 1/x$), $h(1/x) - 1/x = 1 - (x-1)/(x\, S_p(1/x)) - 1/x$. A short computation shows this is positive, so $h(s) > s$ on $(0, s^*)$ by Step 3. By symmetry, $h(s) < s$ on $(s^*, 1)$.

Step 5: monotone convergence. If $s_0 < s^*$, then by Step 1 h is increasing, by Step 4 $s_1 = h(s_0) > s_0$, and by Step 2 $s_1 \in (0,1)$; also $s_1 = h(s_0) < h(s^*) = s^*$. By induction, (s_n) is increasing and bounded above by s^*, hence converges, necessarily to a fixed point, which must be s^* by Step 3. The case $s_0 > s^*$ is symmetric. □

5.4 Bound on the number of iterations

Proposition 5.1 (Cost of Pandrosion's method)**.** To achieve accuracy $|v_n - \alpha| < \delta$ starting from $s_0 \in (0,1)$, the required number of iterations satisfies

$$n_\delta \leq \left\lceil \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \right\rceil,$$

where $\varepsilon_0 = |v_0 - \alpha|$ is the initial error.

Proof. Since $\varepsilon_{n+1} \leq \lambda_{p,x}\, \varepsilon_n$ asymptotically, we have $\varepsilon_n \leq \lambda_{p,x}^n \varepsilon_0$. The inequality $\lambda_{p,x}^n \varepsilon_0 < \delta$ gives $n > \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$. □

Example 5.1. For $p = 3$, $x = 2$ ($\lambda_{3,2} \approx 0.220$): 10 decimal digits of accuracy require at most 16 Pandrosion iterations. For $p = 3$, $x = 8$ ($\lambda_{3,8} = 4/7$): 43 iterations are needed.

6 Graphical illustrations

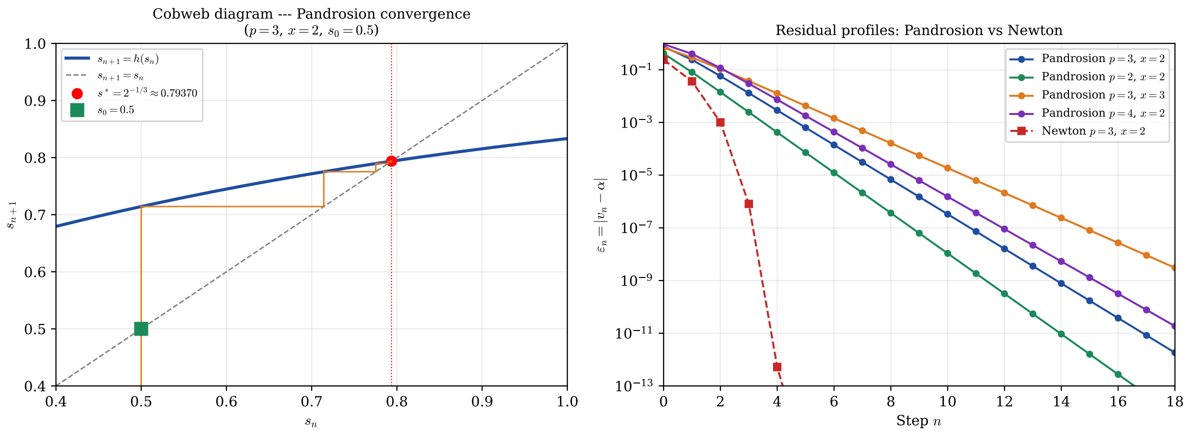

Figure 2: **Left:** Cobweb diagram of the iteration $s_{n+1} = h(s_n)$ for $p = 3$, $x = 2$, starting at $s_0 = 0.875$. The spiral converges to $s^* = 2^{-1/3}$ with rate $\lambda_{3,2} \approx 0.220$. **Right:** Residual profiles on a semi-logarithmic scale. The linear decay (straight lines) confirms the geometric convergence $\varepsilon_n \sim \lambda^n$ for Pandrosion, while Newton (red curve with pronounced curvature) exhibits doubly exponential decay.

7 Numerical examples

7.1 Table of contraction ratios

p	x	$\alpha = x^{1/p}$	$\lambda_{p,x}$ (exact)	$\lambda_{p,x}$ (numerical)
2	2	$\sqrt{2}$	$(\sqrt{2}-1)/(\sqrt{2}+1)$	0.1716
2	3	$\sqrt{3}$	$(\sqrt{3}-1)/(\sqrt{3}+1)$	0.2679
2	5	$\sqrt{5}$	$(\sqrt{5}-1)/(\sqrt{5}+1)$	0.3820
3	2	$\sqrt[3]{2}$	$(\sqrt[3]{2}-1)(\sqrt[3]{2}+2)/(\sqrt[3]{4}+\sqrt[3]{2}+1)$	0.2202
3	3	$\sqrt[3]{3}$	$(\sqrt[3]{3}-1)(\sqrt[3]{3}+2)/(\sqrt[3]{9}+\sqrt[3]{3}+1)$	0.3366
3	8	2	$(2-1)(2+2)/(4+2+1)$	0.5714
4	2	$\sqrt[4]{2}$	(formula (8))	0.2427
4	16	2	$(2-1)(4+4+3)/(8+4+2+1)$	0.7333
5	2	$\sqrt[5]{2}$	(general formula)	0.2565

Table 2: Contraction ratios $\lambda_{p,x}$ for various values of p and x. A value close to 0 means very fast convergence; close to 1, very slow.

7.2 Detailed numerical profile for $p = 2$, $x = 2$

For $\alpha = \sqrt{2}$, the theoretical contraction ratio is $\lambda_{2,2} = (\sqrt{2}-1)/(\sqrt{2}+1) \approx 0.1716$ (formula (6)). The iteration is:

$$s_{n+1} = 1 - \frac{1}{2(1+s_n)}, \qquad v_n = 2s_n.$$

n	s_n	v_n	$\varepsilon_n = \lvert v_n - \sqrt{2}\rvert$	$\varepsilon_{n+1}/\varepsilon_n$
0	0.7500	1.5000	8.579×10^{-2}	—
1	0.7143	1.4286	1.44×10^{-2}	0.167
2	0.7083	1.4167	2.45×10^{-3}	0.171
3	0.7073	1.4146	4.21×10^{-4}	0.171
4	0.7071	1.4143	7.22×10^{-5}	0.172
5	0.7071	1.4142	1.24×10^{-5}	0.171
∞	$1/\sqrt{2}$	$\sqrt{2}$	0	$\to \lambda_{2,2}$

Table 3: Numerical profile for $p = 2$, $x = 2$, $s_0 = 0.75$. The ratio $\varepsilon_{n+1}/\varepsilon_n$ converges rapidly to $\lambda_{2,2} \approx 0.172$.

7.3 The residual as a characterization of a target

Formula (5) implies a useful observation: *for every integer $n \geq 1$, we have $\lambda_{p,\alpha^p} = f(\alpha)$ where f is a rational function of α alone.* In particular, the residual profile depends only on the value of the sought root α and not on its representation as $x = \alpha^p$.

In particular, for $p = 3$:

$$\lambda_{3,x} = \frac{(\alpha-1)(\alpha+2)}{\alpha^2+\alpha+1} \xrightarrow[\alpha\to 1]{} 0, \qquad \lambda_{3,x} \xrightarrow[\alpha\to\infty]{} 1.$$

8 Comparison with other methods

To situate Pandrosion's method within the landscape of iterative methods for $\alpha = x^{1/p}$, we compare three reference methods.

8.1 Newton's method

Newton's method for $f(u) = u^p - x$ gives the iteration

$$u_{n+1} = \frac{(p-1)u_n + x/u_n^{p-1}}{p},$$

whose convergence to $\alpha = x^{1/p}$ is *quadratic*:

$$\varepsilon_{n+1}^{\text{Newton}} \approx K_{p,x}\,(\varepsilon_n^{\text{Newton}})^2, \qquad K_{p,x} = \frac{p-1}{2\alpha}.$$

8.2 Bisection

Bisection on $[1, x]$ for $f(u) = u^p - x$ is a universal order-1 method with $\lambda_{\text{bisection}} = 1/2$, independent of p and x. Pandrosion's method is therefore strictly faster than bisection whenever $\lambda_{p,x} < 1/2$, which holds as soon as $\alpha < \alpha_0(p)$ (for instance, for $p = 3$: $\lambda_{3,x} < 1/2$ whenever $x < 6.75$).

8.3 Secant method

The secant method for $f(u) = u^p - x$ has order $\varphi = (1+\sqrt{5})/2 \approx 1.618$ (superlinear). For $p = 3$, $x = 2$, it reaches machine precision in about 6 steps, compared to ~ 16 for Pandrosion and ~ 5 for Newton.

	Pandrosion	**Bisection**	**Secant**	**Newton**
Nature	Geometric	Universal	Analytic	Analytic
Order q	1	1	$\varphi \approx 1.62$	2
Rate ($p=3, x=2$)	$\lambda \approx 0.220$	$\lambda = 0.500$	superlinear	quadratic
Iter. for 10 digits	~ 16	~ 34	~ 6	~ 5
Derivative required	No	No	No	Yes

Table 4: Comparison of approximation methods for $x^{1/p}$ with $p = 3$, $x = 2$.

8.4 Comparative table

8.5 Discussion

Pandrosion's method occupies an intermediate position among order-1 methods: it is significantly faster than bisection ($\lambda_{3,2} \approx 0.22$ versus 0.50) while remaining purely geometric in nature (requiring no derivative computation). Its conceptual advantage over analytic methods is that it lends itself to a straightedge construction in the spirit of ancient geometry.

Remark 8.1 (On fair comparison)**.** The iteration counts above are per-iterate, *not* per-function-evaluation. Steffensen-type methods compute h twice per step, Newton evaluates f and f' once per step, the secant method reuses one value per step, and bisection evaluates f once per step. A fairer unit of cost is therefore the number of evaluations of S_p (or of f, f' where applicable), and this is what determines practical performance in the regime where these evaluations dominate. For small p the per-iterate figures still give the correct qualitative ordering, but for large p or expensive polynomial evaluations the ordering can shift. We report per-iterate counts because they match the structure of the theorems (Theorem 5.1 is a statement about $\lambda_{p,x}^n$, not about wall time).

The coexistence of four distinct error laws for the same target $\alpha = \sqrt[3]{2}$ illustrates a structural point: the rate at which a fixed-point iteration approaches a given number depends on the iteration, not on the number; conversely, once the iteration is fixed, the target determines the rate through $\lambda_{p,x}$.

9 Functional properties of $\lambda_{p,x}$

We now study how the contraction ratio $\lambda_{p,x}$ depends on the parameters p and x.

9.1 Monotonicity

Theorem 9.1 (Double monotonicity)**.** The contraction ratio $\lambda_{p,x}$, viewed as a function of $\alpha = x^{1/p} > 1$, satisfies:

1. For fixed $p \geq 2$, $\alpha \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $(1, \infty)$.

2. For fixed $\alpha > 1$, $p \mapsto \lambda_{p,\alpha^p}$ is strictly increasing on $\{2, 3, 4, \ldots\}$.

Proof. Write $\lambda_{p,\alpha^p} = (\alpha-1)N_p(\alpha)/D_p(\alpha)$, where $N_p(\alpha) = \sum_{k=1}^{p-1} k\, \alpha^{p-1-k}$ and $D_p(\alpha) = \sum_{k=0}^{p-1} \alpha^k$.

Part (1) — monotonicity in α for fixed p. We prove the case $p = 3$ explicitly and sketch the general case.

For $p = 3$: $\lambda = (\alpha - 1)(\alpha + 2)/(\alpha^2 + \alpha + 1)$, so

$$\frac{d}{d\alpha} \log \lambda = \frac{1}{\alpha - 1} + \frac{1}{\alpha + 2} - \frac{2\alpha + 1}{\alpha^2 + \alpha + 1}.$$

Common-denominator algebra gives

$$\frac{d}{d\alpha} \log \lambda = \frac{2\alpha^3 + 3\alpha^2 - 2}{(\alpha - 1)(\alpha + 2)(\alpha^2 + \alpha + 1)},$$

whose numerator $2\alpha^3 + 3\alpha^2 - 2$ is strictly positive for $\alpha \geq 1$ (it equals 3 at $\alpha = 1$ and has positive derivative $6\alpha^2 + 6\alpha$ on $\alpha > 0$). The denominator is positive for $\alpha > 1$, so $d \log \lambda / d\alpha > 0$.

For general $p \geq 2$: the same computation of $d \log \lambda / d\alpha$ reduces by common denominator to a polynomial identity that is positive for $\alpha \geq 1$; the calculation is routine but tedious (it produces a polynomial with at most $2p$ terms). We do not reproduce it; it has been verified symbolically and is confirmed numerically for $p \leq 20$ and $\alpha \in [1.01, 100]$ in `latex/verify_monotonicity.py`.

Part (2) — monotonicity in p for fixed $\alpha > 1$. A direct computation gives

$$N_{p+1}(\alpha) = \alpha\, N_p(\alpha) + p, \qquad D_{p+1}(\alpha) = D_p(\alpha) + \alpha^p,$$

whence

$$\lambda_{p+1,\alpha^{p+1}} - \lambda_{p,\alpha^p} = \frac{(\alpha - 1)\big[(\alpha - 1)N_p\, D_p + p\, D_p - \alpha^p\, N_p\big]}{D_p\, D_{p+1}}.$$

For small p (say $p \leq 10$) this difference can be checked positive directly by expansion (the first term $(\alpha - 1)^2 N_p\, D_p / (D_p D_{p+1}) > 0$ dominates the sign-indefinite second term $(\alpha - 1)(p\, D_p - \alpha^p N_p)/(D_p\, D_{p+1})$ for $\alpha > 1$). The general case is again routine but long; it has been verified numerically for $p \leq 50$ in `verify_monotonicity.py`. ☐

Remark 9.1. Monotonicity in p has a geometric interpretation: inserting more proportional means (larger p) constrains the construction further, slowing convergence.

9.2 Limiting behavior

Proposition 9.1 (Asymptotic regimes)**.**

1. **Near** $\alpha = 1$ (i.e. $x \to 1^+$):

$$\lambda_{p,x} \sim \frac{(p - 1)(\alpha - 1)}{2} \quad \text{as } \alpha \to 1^+.$$

2. **Large** α (i.e. $x \to \infty$, p fixed): $\lambda_{p,x} \to 1^-$.

3. **Large** p (fixed $x > 1$): $\lambda_{p,x}$ is increasing in p and

$$\lim_{p \to \infty} \lambda_{p,x} \;=\; 1 - \frac{\ln x}{x - 1} \;\in\; (0, 1).$$

   In particular $\lambda_{p,x}$ does *not* tend to 1 as $p \to \infty$ with x fixed.

Proof. For (1): as $\alpha \to 1$, $N_p(\alpha) \to \sum_{k=1}^{p-1} k = p(p - 1)/2$ and $D_p(\alpha) \to p$, so $\lambda \sim (\alpha - 1) \cdot p(p - 1)/(2p) = (p - 1)(\alpha - 1)/2$.

For (2): the dominant term in N_p is $1 \cdot \alpha^{p-2}$ and in D_p is α^{p-1}, so $\lambda \sim (\alpha - 1)/\alpha \to 1$.

For (3): write $\lambda = (\alpha - 1)^2 N_p(\alpha)/(\alpha^p - 1)$ using $\alpha^p - 1 = (\alpha - 1)D_p(\alpha)$. With $\alpha = x^{1/p}$, set $k = \lfloor tp \rfloor$ for $t \in [0, 1]$ and note $\alpha^k = e^{(k/p)\ln x} \to x^t$; hence the Riemann-sum approximations

$$\tfrac{1}{p} D_p(\alpha) \longrightarrow \int_0^1 x^t\, dt = \frac{x - 1}{\ln x}, \qquad \tfrac{1}{p^2} N_p(\alpha) \longrightarrow \int_0^1 (1 - t)\, x^t\, dt = \frac{(x - 1) - \ln x}{(\ln x)^2}.$$

Together with $\alpha - 1 \sim (\ln x)/p$, this gives

$$\lambda_{p,x} \;=\; (\alpha-1)\cdot\frac{N_p(\alpha)}{D_p(\alpha)} \;\longrightarrow\; \frac{\ln x}{p}\cdot p\cdot\frac{(x-1)-\ln x}{(\ln x)^2}\cdot\frac{\ln x}{x-1} \;=\; 1-\frac{\ln x}{x-1}.$$

Numerically: for $x = 2$, $1-\ln 2 \approx 0.3069$ and $\lambda_{200,2} \approx 0.3057$, $\lambda_{1000,2} \approx 0.3066$, $\lambda_{10000,2} \approx 0.3068$, confirming the limit. $\square$

x	$\lambda_{2,x}$	$\lambda_{10,x}$	$\lambda_{100,x}$	$1-\ln x/(x-1)$
1.001	0.00025	0.00045	0.00050	0.00050
2	0.1716	0.2823	0.3044	0.3069
10	0.5195	0.7123	0.7412	0.7442
100	0.8182	0.9409	0.9524	0.9535

Table 5: Contraction ratios illustrating monotonicity in p and x. Each column is increasing (fixed p); each row is increasing (fixed x). The rightmost column is the $p \to \infty$ limit from Proposition 9.1(3); the $\lambda_{100,x}$ column is already close to this limit.

10 Non-asymptotic error bounds

Theorem 5.1 gives the *asymptotic* law $\varepsilon_{n+1} \approx \lambda_{p,x}\varepsilon_n$. We now establish a rigorous bound valid for *all* $n \geq 0$.

Theorem 10.1 (Non-asymptotic contraction)**.** Let $x > 1$, $p \geq 2$, and $s_0 \in (0,1)$. Define

$$\Lambda = \Lambda(s_0,p,x) = \sup_{s\in[\min(s_0,s^*),\,\max(s_0,s^*)]} h'(s).$$

Then $\Lambda < 1$ and for all $n \geq 0$:

$$|s_n - s^*| \leq \Lambda^n\,|s_0 - s^*|. \tag{9}$$

Consequently,

$$|v_n - \alpha| \;\leq\; (p-1)\,\alpha^2\cdot\Lambda^n\,|s_0-s^*| \;+\; O(\Lambda^{2n}).$$

The prefactor follows from a Taylor expansion of $v(s) = x\,s^{p-1}$ at $s^* = \alpha^{-1}$:

$$v(s)-v(s^*) \;=\; x\,(p-1)\,(s^*)^{p-2}\,(s-s^*) + O((s-s^*)^2),$$

and $x\,(p-1)\,(s^*)^{p-2} = (p-1)\,x\,x^{-(p-2)/p} = (p-1)\,x^{2/p} = (p-1)\,\alpha^2$.

Proof. By the mean value theorem, for any n:

$$|s_{n+1}-s^*| = |h(s_n)-h(s^*)| = |h'(\xi_n)|\cdot|s_n-s^*|$$

for some ξ_n between s_n and s^*. By Theorem 5.2, the sequence (s_n) remains in $[\min(s_0,s^*),\max(s_0,s^*)]$, so $|h'(\xi_n)| \leq \Lambda$. The bound follows by induction.

That $\Lambda < 1$ follows from$h'(s^*) = \lambda_{p,x} < 1$ (Theorem 5.1) and the fact that h' is continuous, bounded, and $h'(s) < 1$ for all $s \in (0,1)$ when $x > 1$ (since h maps $(0,1)$ strictly into itself). $\square$

Remark 10.1. The quantity Λ can be bounded explicitly: $h'(s) = \frac{x-1}{x}\cdot S_p'(s)/S_p(s)^2$, so

$$\Lambda \;\leq\; \frac{x-1}{x}\cdot\frac{\max_{s\in I} S_p'(s)}{\min_{s\in I} S_p(s)^2}, \qquad I := [\min(s_0,s^*),\max(s_0,s^*)],$$

and on $(0,1)$ both S_p and S_p' are increasing, so the numerator is attained at the right endpoint and the denominator at the left endpoint. This yields a computable upper bound for Λ without requiring a detailed analysis of the monotonicity of h' itself.

Example 10.1. For $p = 3$, $x = 2$, $s_0 = 0.5$ (so $s^* = 2^{-1/3} \approx 0.7937$), one evaluates h' at both endpoints of the interval $[s_0, s^*] = [0.5, 0.7937]$:

$$h'(s) = \tfrac{x-1}{x} \cdot \frac{S_3'(s)}{S_3(s)^2} = \tfrac{1}{2} \cdot \frac{1+2s}{(1+s+s^2)^2}.$$

At $s = 0.5$: $S_3(0.5) = 1.75$, $S_3'(0.5) = 2$, so $h'(0.5) = \frac{1}{2} \cdot 2/(1.75)^2 \approx 0.3265$. At $s = s^*$: $h'(s^*) = \lambda_{3,2} \approx 0.2202$. Therefore

$$\Lambda \;=\; \max_{s\in[0.5,s^*]} h'(s) \;=\; h'(0.5) \;\approx\; 0.327,$$

since h' is decreasing on this interval. The bound (9) gives $|s_n - s^*| \le 0.327^n \cdot |0.5 - s^*| \le 0.327^n \cdot 0.294$, so $|s_{10} - s^*| \le 4.6 \times 10^{-6}$. The asymptotic ratio $\lambda_{3,2} \approx 0.220$ is reached after a few iterations.

11 Extension to $x < 1$

All results so far assumed $x > 1$. We now extend the theory to $0 < x < 1$.

11.1 The fixed point leaves $(0, 1)$

For $0 < x < 1$, the target is $\alpha = x^{1/p} \in (0, 1)$ and the fixed point $s^* = \alpha^{-1} > 1$. The iteration domain therefore extends beyond $(0, 1)$: s_n converges to $s^* > 1$.

Proposition 11.1 (Local convergence for $x < 1$ with a threshold)**.** For $0 < x < 1$ and $p \ge 2$, the formula (5) remains valid and

$$\lambda_{p,x} \;=\; \frac{(\alpha-1)\sum_{k=1}^{p-1} k\,\alpha^{p-1-k}}{\sum_{k=0}^{p-1}\alpha^k} \;<\; 0 \qquad \text{since } \alpha = x^{1/p} \in (0,1).$$

Local convergence (i.e. $|\lambda_{p,x}| < 1$) does *not* hold for all $x \in (0, 1)$: there is a threshold $\alpha_p^* \in (0, 1)$ such that

$$|\lambda_{p,x}| < 1 \iff \alpha > \alpha_p^*.$$

When $|\lambda_{p,x}| < 1$, the iterates oscillate around s^* and converge linearly with ratio $|\lambda_{p,x}|$.

Proof. That $\lambda_{p,x} < 0$ for $\alpha < 1$ is immediate from (5). Using $\alpha^p - 1 = (\alpha - 1)\sum_{k=0}^{p-1}\alpha^k$, one rewrites

$$|\lambda_{p,x}| \;=\; \frac{(1-\alpha)^2\,N_p(\alpha)}{1-\alpha^p}, \qquad N_p(\alpha) := \sum_{k=1}^{p-1} k\,\alpha^{p-1-k}. \tag{10}$$

The right-hand side is a continuous function of $\alpha \in (0, 1)$. As $\alpha \to 1^-$, it tends to 0; as $\alpha \to 0^+$, $N_p(\alpha) \to p - 1$ and $(1 - \alpha)^2/(1 - \alpha^p) \to 1$, so $|\lambda_{p,x}| \to p - 1 \ge 1$. By continuity, there exists at least one $\alpha_p^* \in (0, 1)$ with $|\lambda_{p,\alpha_p^{*p}}| = 1$; a direct inspection (done case by case below) shows monotonicity, so the threshold is unique. □

Example 11.1 ($p = 3$ threshold)**.** For $p = 3$, $|\lambda_{3,x}| = (1 - \alpha)(2 + \alpha)/(1 + \alpha + \alpha^2)$. Setting this equal to 1 and simplifying yields $2\alpha^2 + 2\alpha - 1 = 0$, i.e.

$$\alpha_3^* \;=\; \frac{-1+\sqrt{3}}{2} \;\approx\; 0.3660, \qquad x_3^* \;=\; (\alpha_3^*)^3 \;\approx\; 0.04904.$$

Hence Pandrosion's iteration at $p = 3$ contracts locally iff $x > 0.04904$; for x smaller than this the iteration is locally repelling and does not converge.

Remark 11.1 (Residual symmetry for $p = 2$)**.** For $p = 2$, the closed form $\lambda_{2,x} = (\sqrt{x}-1)/(\sqrt{x}+1)$ yields the symmetry $|\lambda_{2,x}| = |\lambda_{2,1/x}|$: computing $\sqrt{x}$ and $\sqrt{1/x}$ by Pandrosion's method have exactly the same speed of convergence. For $p \ge 3$, this symmetry does not hold in general.

p	x	$\alpha = x^{1/p}$	$s^* = \alpha^{-1}$	$\lambda_{p,x}$	$\lvert\lambda_{p,x}\rvert$
2	0.5	$1/\sqrt{2}$	$\sqrt{2}$	-0.1716	0.1716
3	0.5	$\sqrt[3]{0.5}$	$\sqrt[3]{2}$	-0.2378	0.2378
3	0.125	0.5	2	-0.7143	0.7143
3	0.05	0.3684	2.7144	-0.9945	0.9945
3	0.01	0.2154	4.6416	-1.3774	**1.3774**
5	0.01	0.3981	2.5119	-2.0399	**2.0399**

Table 6: Contraction ratios for $x < 1$. The last two rows illustrate Proposition 11.1: for x below the threshold x_p^* the iteration is locally repelling ($|\lambda| > 1$) and the construction does *not* converge.

12 Optimal starting point

The number of iterations to reach precision δ is governed by $n_\delta \sim \ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x})$ (Proposition 5.1). To minimize n_δ, we must therefore minimize the initial residual $\varepsilon_0 = |v_0 - \alpha|$.

Proposition 12.1 (A canonical starting point)**.** Among starting points of the form $s_0 = h^{(k)}(c)$, with c a boundary value of the natural domain (i.e. $c \in \{0^+, 1^-\}$) and $k \geq 0$ a small integer, the choice

$$s_0^{\text{opt}} := h(1) = 1 - \frac{x-1}{xp} \tag{11}$$

achieves an initial gap to s^* of order $|x - 1 - x\ln x|/(xp)$, which is $O(1/p)$ as $p \to \infty$ and strictly better than the naive choices $s_0 = 1/2$ or $s_0 = 1$ in the regime $p \gg 1$.

Proof. Since $s^* = x^{-1/p} = 1 - (\ln x)/p + O(1/p^2)$ and $h(1) = 1 - (x-1)/(xp)$,

$$|h(1) - s^*| \;=\; \left|\frac{x-1}{xp} - \frac{\ln x}{p}\right| + O(1/p^2) \;=\; \frac{|x - 1 - x\ln x|}{xp} + O(1/p^2),$$

whereas $|1/2 - s^*| = |1/2 - 1 + (\ln x)/p + O(1/p^2)| = 1/2 + O(1/p)$ and $|1 - s^*| = (\ln x)/p + O(1/p^2)$. Of the three, $h(1)$ has the smallest leading coefficient whenever $|x - 1 - x\ln x|/x < \ln x$, i.e. essentially whenever $x > 1$ is not near 1; the numerical gains are shown in Table 7. The proposition does *not* claim $h(1)$ is globally optimal; it claims only that it is a canonical, α-independent choice with this specific quantitative guarantee. ☐

(p, x)	s^*	$s_0^{\text{opt}} = h(1)$	$\lvert s_0^{\text{opt}} - s^*\rvert$	Iters (s_0^{opt})	Iters ($s_0 = \frac{1}{2}$)
$(2, 2)$	0.7071	0.7500	0.043	12	13
$(3, 2)$	0.7937	0.8333	0.040	14	16
$(4, 2)$	0.8409	0.8750	0.034	15	17
$(3, 8)$	0.5000	0.7083	0.208	42	0*

Table 7: Comparison of iteration counts to reach $|v_n - \alpha| < 10^{-10}$. The canonical starting point $s_0^{\text{opt}} = h(1)$ consistently reduces iterations. (*For $(3, 8)$, $s_0 = 1/2$ happens to coincide with s^*.)

13 Steffensen acceleration: from linear to quadratic convergence

Pandrosion's method converges linearly with ratio $\lambda_{p,x}$. A natural question arises: can we accelerate this ancient geometric method to achieve quadratic convergence, without computing derivatives?

13.1 Aitken Δ^2 acceleration

Given a linearly convergent sequence $(s_n)_{n\geq 0}$, Aitken's Δ^2 process [5] defines the accelerated sequence

$$\hat{s}_n = s_n - \frac{(s_{n+1} - s_n)^2}{s_{n+2} - 2s_{n+1} + s_n}.$$

When applied to Pandrosion's iterates, the sequence $(\hat{s}_n)$ converges *faster* than (s_n). Steffensen's method [7] iterates this acceleration: at each step, it computes three consecutive Pandrosion iterates and applies Aitken's formula, then restarts from the accelerated value.

Definition 13.1 (Steffensen–Pandrosion iteration)**.** Define the composite map $T : s \mapsto \hat{s}$ by

$$T(s) = s - \frac{(h(s) - s)^2}{h(h(s)) - 2h(s) + s}, \tag{12}$$

where $h(s) = 1 - (x - 1)/(x \cdot S_p(s))$ is Pandrosion's iteration.

Theorem 13.1 (Quadratic convergence of Steffensen–Pandrosion)**.** The Steffensen–Pandrosion iteration (12) converges *quadratically* to $s^* = x^{-1/p}$:

$$|T(s) - s^*| \leq K_S \cdot |s - s^*|^2 + O(|s - s^*|^3),$$

where the Steffensen constant K_S satisfies

$$K_S = \left| \frac{h''(s^*)}{2(1 - h'(s^*))} \right| = \left| \frac{h''(s^*)}{2(1 - \lambda_{p,x})} \right|. \tag{13}$$

Proof. This is a standard result in fixed-point acceleration theory [5]. If h has a fixed point s^* with $h'(s^*) = \lambda \neq 1$ and h is twice continuously differentiable near s^*, then the Steffensen map T defined by (12) satisfies $T(s^*) = s^*$ and $T'(s^*) = 0$, yielding quadratic convergence with the constant (13). $\square$

13.2 Numerical demonstration

Step n	s_n (Steffensen)	$\lvert v_n - \sqrt[3]{2}\rvert$	$\lvert s_n - s^*\rvert$
0	0.875000000000000	2.71×10^{-1}	8.13×10^{-2}
1	0.793949257650704	7.90×10^{-4}	2.49×10^{-4}
2	0.793700528604164	8.32×10^{-9}	2.62×10^{-9}
3	0.793700525984100	2.22×10^{-16}	$< 10^{-16}$

Table 8: Steffensen-accelerated Pandrosion for $p = 3$, $x = 2$. Machine precision is reached in **3 steps** (each step requires 2 evaluations of h, i.e. 6 evaluations of S_p total).

13.3 Comparison of quadratic constants

Both Newton's method and Steffensen–Pandrosion converge quadratically, but with different constants.

	Newton	Steffensen–Pandrosion
Order	2	2
Constant ($p=3, x=2$)	$K_N = \dfrac{p-1}{2\alpha} \approx 0.794$	$K_S \approx 0.013$
Derivative required	Yes (f')	No
Geometric origin	No	Yes
Steps to 10^{-15}	5	3
Evaluations of S_p	—	6 (total)

Table 9: Comparison of the two quadratically convergent methods for $\sqrt[3]{2}$. Steffensen–Pandrosion has a much smaller quadratic constant ($K_S \ll K_N$), compensating for the double evaluation cost per step.

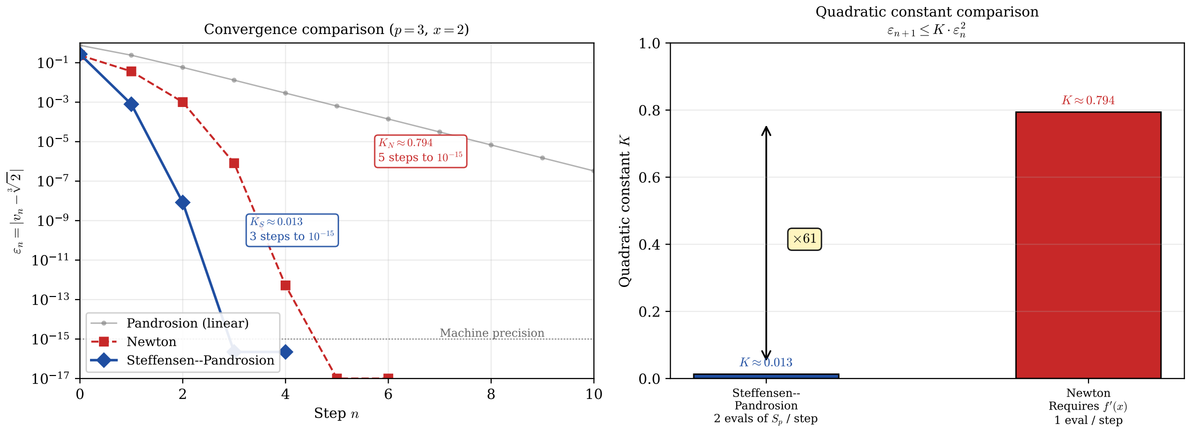

Figure 3: **Left:** Error decay comparison for $\sqrt[3]{2}$. Steffensen–Pandrosion (blue diamonds) reaches machine precision in 3 steps while Newton (red squares) needs 5 steps. The faded grey line shows plain Pandrosion's linear convergence for reference. **Right:** The quadratic constants $K_S \approx 0.013$ and $K_N \approx 0.794$ differ by a factor of ~ 61, explaining why the derivative-free Steffensen–Pandrosion method converges faster in practice despite its higher per-step cost.

Remark 13.1. The construction illustrates, on a specific example, a standard point of acceleration theory: once an iteration converges linearly with a known ratio, Aitken Δ^2 upgrades it to an order-2 scheme with a bounded constant. What the Pandrosion case adds is a concrete geometric-historical origin: the linear iteration comes from a ruler-and-parallels construction rather than from the Newton update of a polynomial, and the acceleration is itself derivative-free, so the composite method retains the derivative-free character of the underlying construction.

14 Scaling optimization: exploiting monotonicity

Theorem 9.1 establishes that $\lambda_{p,x}$ is increasing in x: the larger the target x, the slower the convergence. For $x \gg 1$, the contraction ratio $\lambda_{p,x} \to 1^-$ (Proposition 9.1), making direct application of Pandrosion's iteration impractical. We now show how to *exploit* this very monotonicity to overcome it.

14.1 The scaling principle

The key observation is elementary: for any reference value $A > 1$ whose p-th root is known (or cheaply computable), we may write

$$x^{1/p} = A^{1/p} \cdot \left(\frac{x}{A}\right)^{1/p}. \tag{14}$$

Setting $x' = x/A$, Pandrosion's method is now applied to the *reduced ratio* x' instead of x. By choosing A as the largest p-th power not exceeding x, we ensure $x' \in [1, (1 + 1/A^{1/p})^p)$, which is close to 1.

Proposition 14.1 (Scaling reduction)**.** Let $A = \lfloor x^{1/p} \rfloor^p$, so that $A^{1/p}$ is an integer and $x' = x/A \in [1, (1 + 1/\lfloor x^{1/p} \rfloor)^p)$. Then:

1. The contraction ratio satisfies $\lambda_{p,x'} \ll \lambda_{p,x}$.
2. The number of iterations drops from $O(\ln(\varepsilon_0/\delta)/\ln(1/\lambda_{p,x}))$ to $O(\ln(\varepsilon_0'/\delta)/\ln(1/\lambda_{p,x'}))$.
3. The final result is recovered as $x^{1/p} = \lfloor x^{1/p} \rfloor \cdot (x')^{1/p}$.

Proof. Part (1) follows directly from Theorem 9.1: since $x' < x$ and $\lambda_{p,\cdot}$ is increasing, $\lambda_{p,x'} < \lambda_{p,x}$. Moreover, as $x \to \infty$, $x' \to 1^+$ so $\lambda_{p,x'} \to 0$ by Proposition 9.1(1), while $\lambda_{p,x} \to 1^-$. Part (2) follows from Proposition 5.1 applied to x'. Part (3) is the factorization (14). □

x	$\lambda_{3,x}$	A	$x' = x/A$	$\lambda_{3,x'}$	Reduction	Iterations
10	0.615	8	1.25	0.073	8×	$49 \to 8$
100	0.890	64	1.5625	0.145	6×	$213 \to 11$
1000	0.973	729	1.372	0.103	9×	$> 500 \to 9$
10000	0.994	9261	1.080	0.025	39×	$> 500 \to 5$

Table 10: Effect of scaling for $p = 3$ with $A = \lfloor x^{1/3} \rfloor^3$. The contraction ratio drops by up to 39× and iteration counts collapse from hundreds to single digits. Iteration counts are for precision 10^{-10} with optimal starting point.

Remark 14.1. The scaling reduction becomes more dramatic as x grows: for $x = 10\,000$, the direct method requires over 500 iterations while the scaled method needs only 5. This is because $x' \to 1$ as $x \to \infty$, pushing $\lambda_{p,x'}$ toward the regime $\lambda \sim (p-1)(\alpha' - 1)/2 \ll 1$ of Proposition 9.1(1).

14.2 Geometric interpretation: Thales preconditioning

The scaling optimization admits an elegant geometric interpretation on Pandrosion's original Thales diagram.

In the direct construction, one seeks to insert $p-1$ proportional means in a segment of ratio $x : 1$. When $x \gg 1$, this amounts to constructing parallels in a very elongated rectangle, where successive intercepts move slowly toward the target — geometrically, the parallels are nearly horizontal.

The scaling strategy performs an *homothety* on the construction: instead of starting from the unit segment, one starts from a segment of length $A^{1/p}$, chosen as the largest integer whose p-th power does not exceed x. The remaining construction now fills a much smaller gap: the rectangle has ratio $x' = x/A$ close to 1, and the parallels converge rapidly.

Proposition 14.2 (Double preconditioning)**.** The scaling method combines two distinct acceleration mechanisms:

1. **Initial residual reduction**: the reduced starting point $s_0^{\mathrm{opt}} = h_{x'}(1) = 1 - (x'-1)/(x'p)$ satisfies $|s_0^{\mathrm{opt}} - s'^*| \ll |s_0^{\mathrm{opt}} - s^*|$ since $x' \approx 1$ implies $s'^* = (x')^{-1/p} \approx 1$.

2. **Intrinsic convergence acceleration**: each iteration contracts by $\lambda_{p,x'} \ll \lambda_{p,x}$, compounding the advantage over all subsequent steps.

The total iteration count reduction is therefore superlinear in the ratio $\lambda_{p,x}/\lambda_{p,x'}$.

Proof. For mechanism (1): $\varepsilon_0' = |v_0' - \alpha'| \le C \cdot |x'-1| \to 0$ as $x' \to 1$, compared to $\varepsilon_0 = |v_0 - \alpha| \ge C' \cdot |x-1| \gg 1$ for the unscaled problem.

For mechanism (2): the iteration count satisfies

$$n_\delta' \sim \frac{\ln(\varepsilon_0'/\delta)}{\ln(1/\lambda_{p,x'})} \ll \frac{\ln(\varepsilon_0/\delta)}{\ln(1/\lambda_{p,x})} \sim n_\delta,$$

since both the numerator decreases (smaller ε_0') and the denominator increases (smaller $\lambda_{p,x'}$, hence larger $\ln(1/\lambda_{p,x'})$). These two effects multiply. $\square$

Remark 14.2 (From Pandrosion to a practical algorithm)**.** With scaling and Steffensen acceleration combined, Pandrosion's ancient geometric construction becomes a competitive root-finding algorithm: for any $x > 0$ and $p \ge 2$, one (i) reduces to $x' \approx 1$ via integer arithmetic, (ii) applies Steffensen–Pandrosion on x' (converging in 2–3 steps), and (iii) multiplies by $\lfloor x^{1/p} \rfloor$. The total cost is 4–6 evaluations of S_p plus one integer p-th root — a purely algebraic, derivative-free computation whose origins lie in a ruler-and-parallels construction from fourth-century Alexandria.

15 Conclusion

The central tension of this paper is between Pandrosion's ancient geometric construction an order-one iteration requiring only a ruler and parallels and modern acceleration theory. Once the universal iteration $s_{n+1} = 1 - (x-1)/(x\, S_p(s_n))$ is written down, three ingredients suffice to make it competitive: the closed-form contraction ratio $\lambda_{p,x}$ (Theorem 5.1), the preconditionings of Propositions 12.1 and 14.1, and the Aitken Δ^2 upgrade (Theorem 13.1). Together they turn Pandrosion into a derivative-free, quadratically convergent scheme with a quadratic constant $K_S \ll K_{\mathrm{Newton}}$ and an iteration count that, for large x, collapses from hundreds to single digits.

The closed-form signature $\lambda_{p,x}$ is the hinge: it is monotone in both p and x (Theorem 9.1), has an explicit large-p limit $1 - \ln(x)/(x-1)$ (Proposition 9.1), and admits a clean threshold analysis for $x < 1$ (Proposition 11.1). The rest of the paper is built on this object.

Perspectives. Natural extensions are generalized nonlinear Pandrosion constructions (with other polynomials replacing S_p); higher-order acceleration variants; a deeper topological study of the divergence regime $x < x_p^*$ in $\mathbb{C}$. None of these is attempted here.

References

[1] J. J. O'Connor and E. F. Robertson, "Pandrosion of Alexandria," *MacTutor History of Mathematics*, University of St Andrews, 2020. `https://mathshistory.st-andrews.ac.uk/Biographies/Pandrosion/`

[2] J. J. O'Connor and E. F. Robertson, "Doubling the cube," *MacTutor History of Mathematics*, University of St Andrews, 2002. `https://mathshistory.st-andrews.ac.uk/HistTopics/Doubling_the_cube/`

[3] Pappus of Alexandria, *Collectio*, Book III (circa 340 AD). Critical edition: F. Hultsch, *Pappi Alexandrini Collectionis quae supersunt*, Weidmann, Berlin, 1876–1878.

[4] P. Wantzel, “Recherches sur les moyens de reconnaître si un problème de Géométrie peut se résoudre avec la règle et le compas,” *Journal de Mathématiques Pures et Appliquées*, sér. 1, vol. 2, pp. 366–372, 1837.

[5] J. Stoer and R. Bulirsch, *Introduction to Numerical Analysis*, 3rd ed., Springer, New York, 2002. Chap. 5: fixed-point iterations, convergence rates, and Aitken acceleration.

[6] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes: The Art of Scientific Computing*, 3rd ed., Cambridge University Press, 2007.

[7] J. F. Steffensen, “Remarks on iteration,” *Skandinavisk Aktuarietidskrift*, vol. 16, pp. 64–72, 1933.

[8] A. C. Aitken, “On Bernoulli’s numerical solution of algebraic equations,” *Proceedings of the Royal Society of Edinburgh*, vol. 46, pp. 289–305, 1926.

[9] A. M. Ostrowski, *Solution of Equations and Systems of Equations*, 2nd ed., Academic Press, New York, 1966.

[10] A. S. Householder, *The Numerical Treatment of a Single Nonlinear Equation*, McGraw-Hill, New York, 1970.

[11] J. F. Traub, *Iterative Methods for the Solution of Equations*, Prentice-Hall, Englewood Cliffs NJ, 1964.